

Adonay Tesfamichael

IT214-006

Happy learning

This diagram shows a school structure. It is designed to show how the school operates and has all the information from the students, school, teachers and parents.



Conceptual Design:

For this diagram the main entities we are creating are the STUDENT, TEACHER, CLASS, PARENT. For our first relationship we have the TEACHER_AT to CLASS_AT Which is many to many (M:M) because one teacher can be in many classes and a class can have many teachers and this relationship is held by a bridge called ASSIGNMENT_AT, which holds FK: CLASS_AT_CODE_AT and TEACHER_AT_ID_AT. Next one is the PARENT_AT to STUDENT_AT which is a many to many (M:M) relationship because students can have more than one parent and parents can have more than one kid and this relationship is held by a bridge called RELATIONSHIP, which is holding FK: STUDENT_AT_ID_AT and PARENT_AT_ID_AT. We also have the STUDENT_AT to CLASS_AT which is many to one (M:1) because a class can have many kids and kids can only be in one class time. We also have three new entities LUNCH_ORDER_AT, LUNCH_ITEM_AT and ORDER_ITEM_AT. LUNCH_ORDER_AT has a (1:M) relationship to the STUDENT_AT with a FK: STUDENT_AT_ID_AT. Last relation is between LUNCH_ORDER_AT and LUNCH_ITEM_AT that is M:M which has a bridge called ORDER_ITEM_AT.

Entities Analysis:

- STUDNET_AT

- This entity shows the students information table.
- For this entity since there is no PK given we will use a surrogate key ID_AT and the foreign key is CLASS_AT_CODE_AT which comes from CLASS_AT ID CODE_AT.

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
ID_AT	INT	Simple	Single	Required	Unique
FIRST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
MIDDLE_NAME_AT	VARCHAR(15)	Simple	Single	Optional	Duplicated
LAST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
NICK_NAME_AT	VARCHAR(10)	Simple	Single	Required	Duplicated
DATE_OF_BIRTH_AT	DATE	Simple	Single	Required	Duplicated
ADDRESS_AT	VARCHAR(55)	Composite	Single	Required	Duplicated
CITY_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
ZIPCODE_AT	DECIMAL(0,5)	Simple	Single	Required	Duplicated
STREET_AT	VARCHAR(35)	Simple	Single	Required	Duplicated
CLASS_AT_CODE_AT	VARCHAR(2)	Simple	Single	Required	Duplicated

- PARENT_AT

- This entity shows the parent/guardian information table.
- For this entity since there is no PK given, we will use a surrogate key ID_AT and there is no foreign key

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
ID_AT	INT	Simple	Single	Required	Unique
FIRST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
LAST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
MIDDLE_NAME_AT	VARCHAR(15)	Simple	Single	Optional	Duplicated
HOME_PHONE_AT	CHAR(13)	Simple	Single	Optional	Duplicated
MOBILE_PHONE_AT	CHAR(13)	Simple	Single	Required	Duplicated
WORK_PHONE_AT	CHAR(13)	Simple	Single	Optional	Duplicated
EMAIL_AT	VARCHAR(30)	Simple	Single	Optional	Duplicated
RELATION_AT	VARCHAR(20)	Simple	Single	Required	Duplicated

- **CLASS_AT**

- This entity shows the class information table.
- For this entity the primary key is the CODE_AT and there is no foreign key

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
CLASS_CODE_AT	VARCHAR(2)	Composite	Single	Required	Unique
AGE_LEVEL_AT	VARCHAR(7)	Simple	Single	Required	Duplicated
COLOR_AT	VARCHAR(5)	Simple	Single	Required	Duplicated
ROOM_NUMBER_AT	INT	Simple	Single	Required	Duplicated
PHONE_NUMBER_AT	CHAR(13)	Simple	Single	Required	Duplicated
DIRECTION_AT	VARCHAR(160)	Simple	Single	Optional	Duplicated
CAPACITY_AT	INT	Simple	Single	Required	Duplicated

- **TEACHER_AT**

- This entity shows the teacher information table.
- For this entity since there is no PK given, we will use a surrogate key ID_AT and there is no foreign key

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
ID_AT	INT	Simple	Single	Required	Unique
FIRST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
LAST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
MIDDLE_NAME_AT	VARCHAR(15)	Simple	Single	Optional	Duplicated
HOME_PHONE_AT	CHAR(13)	Simple	Single	Required	Duplicated
WORK_EMAIL_AT	VARCHAR(30)	Simple	Single	Required	Duplicated
PERSONAL_EMAIL_AT	VARCHAR(30)	Simple	Single	Optional	Duplicated
COLLEGE_AT	VARCHAR(50)	Simple	Single	Optional	Duplicated
HIGHEST_DEGREE_AT	VARCHAR(5)	Simple	Single	Required	Duplicated
DEGREE_AREA_AT	VARCHAR(25)	Simple	Single	Required	Duplicated

- **RELATIONSHIP_AT**

- This entity shows the relationship between Student & Parent. Since the relationship between Student and Parent is M:M we create a bridge called **RELATIONSHIP** to connect them.

Attribute Name	Domain & Size
STUDENT_AT_ID_AT	INT
PARENT_AT_ID_AT	INT

- **ASSIGNMENT_AT**

- This entity shows the relationship between Class & Teacher. Since the relationship between Class and Teacher is M:M we create a bridge called **ASSIGNMENT** to connect them.

Attribute Name	Domain & Size
CLASS_AT_CODE_AT	INT
TEACHER_AT_ID_AT	INT

- **LUNCH_ORDER_AT**

- This entity shows the Order information table.
- For this entity since there is no PK given, we will use a surrogate key ID_AT and there is a FK: STUDENT_AT_ID_AT references STUDENT_AT(ID_AT)

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
ID_AT	INT	Simple	Single	Required	Unique
QUANTITY_AT	INT	Simple	Single	Required	Duplicated
DATE_AT	DATE	Simple	Single	Required	Duplicated
STUDENT_AT_ID_AT	INT	Simple	Single	Required	Duplicated

- **LUNCH_ITEM_AT**

- This entity shows the item information table.
- we will use a primary key ID_AT, no foreign key

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
CODE_AT	INT	Simple	Single	Required	Unique
ITEM_NAME_AT	VARCHAR(26)	Simple	Single	Required	Duplicated
CALORIES_AT	INT	Simple	Single	Required	Duplicated

- **ORDER_ITEM_AT**

- This entity shows the relationship between **LUNCH_ORDER_AT** & **LUNCH_ITEM_AT**. Since the relationship between LUNCH_ORDER and LUNCH_ITEM is M:M we create a bridge called **ORDER_ITEM_AT** to connect them.

Attribute Name	Domain & Size
LUNCH_ORDER_AT_ID_AT	INT
LUNCH_ITEM_AT_CODE_AT	INT

Relationship Analysis:

→ TEACHER_AT to CLASS_AT

- ◆ We use a bridge : **ASSIGNMENT_AT**
 - TEACHER has class ASSIGNMENT (1:M)
 - CLASS has teacher ASSIGNMENT (1:M)
- ◆ M:M because one teacher can be in many classes and one class can have many teachers.
- ◆ The straight is weak because both entities exist independently, not relying on each other.
- ◆ TEACHER(total) because teachers get assigned to a class and CLASS (total) because all classes have teachers.
- ◆ No special cardinality
- ◆ FK: The ASSIGNMENT table holding the CLASS_AT_CODE_AT reference to the class entity and TEACHER_AT_ID_AT reference to the teacher entity

→ PARENT_AT to STUDENT_AT

- ◆ We use a bridge: **RELATIONSHIP_AT**
 - PARENT has student RELATIONSHIP (1:M)
 - STUDENT has parent RELATIONSHIP (1:M)
- ◆ M:M because students can have more than one parent and parents can have more than one kid
- ◆ The strength is strong because each student is connected to a parent or a caretaker
- ◆ They are both total because they both need each other
- ◆ No special cardinality

- ◆ FK: The RELATIONSHIP table holding the STUDENT_AT_ID_AT reference to the student entity and PARENT_AT_ID_AT reference to the parent entity

→ STUDENT_AT to CLASS_AT

- ◆ STUDENT enrolls in CLASS
- ◆ One CLASS contains many STUDENTS, and one STUDENT is enrolled in one CLASS at a time. No bridge
- ◆ The straight is weak because both entities exist independently, not relying on each other.
- ◆ CLASS (partial) because all the classes might not have kids sometimes, and STUDENTS (total) because a student needs to be always enrolled in a class.
- ◆ No special cardinality
- ◆ FK: CLASS_AT_CODE_AT reference to the CLASS entity

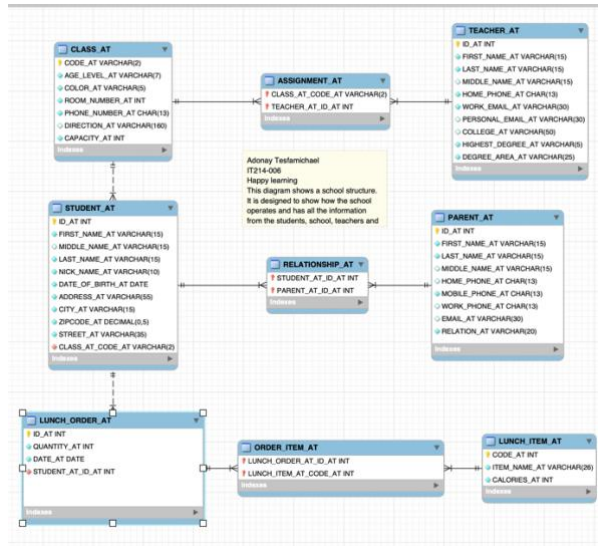
→ STUDENT_AT & LUNCH_ORDER_AT

- ◆ STUDENT_AT orders LUNCH_ORDER_AT
- ◆ 1:M Each student can order one lunch every day, and lunch can be ordered many times every day for many days
- ◆ The straight is weak, because a student might not order lunch.
- ◆ Both is partial because there might not be an order someday to have a menu
- ◆ FK: STUDENT_AT_ID_AT references STUDENT_AT(ID_AT)

→ LUNCH_ORDER_AT & LUNCH_ITEM_AT

- ◆ We use a bridge: ORDER_ITEM_AT
 - LUNCH_ORDER_AT has lunch item ORDER_ITEM_AT(1:M)
 - LUNCH_ITEM_AT has lunch order ORDER_ITEM_AT(1:M)
- ◆ M:M because one order may have up to 4 items and each item will be shows on orders
- ◆ The straight is strong because an order can't be placed without items
- ◆ They are both total because they both need each other
- ◆ FK: The ORDER_ITEM_AT table holding the LUNCH_ORDER_AT_ID_AT reference to the LUNCH_ORDER_AT entity and LUNCH_ITEM_AT_CODE_AT reference to the LUNCH_ITEM_AT entity

Entity Relationship Diagram:



TEACHER_AT

TEACHER_AT - Table					
Name: TEACHER_AT					
Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒	
FIRST_NAME_AT	VARCHAR(15)	☐	☒	☐	
LAST_NAME_AT	VARCHAR(15)	☐	☒	☐	
MIDDLE_NAME_AT	VARCHAR(15)	☐	☐	☐	
HOME_PHONE_AT	CHAR(13)	☐	☒	☐	
WORK_EMAIL_AT	VARCHAR(30)	☐	☒	☐	
PERSONAL_EMAIL_AT	VARCHAR(30)	☐	☐	☐	
COLLEGE_AT	VARCHAR(50)	☐	☐	☐	
HIGHEST_DEGREE_AT	VARCHAR(5)	☐	☒	☐	
DEGREE_AREA_AT	VARCHAR(25)	☐	☒	☐	

STUDENT_AT

STUDENT_AT - Table					
Name: STUDENT_AT					
Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒	
FIRST_NAME_AT	VARCHAR(15)	☐	☒	☐	
MIDDLE_NAME_AT	VARCHAR(15)	☐	☐	☐	
LAST_NAME_AT	VARCHAR(15)	☐	☒	☐	
NICK_NAME_AT	VARCHAR(10)	☐	☒	☐	
DATE_OF_BIRTH_AT	DATE	☐	☒	☐	
ADDRESS_AT	VARCHAR(55)	☐	☒	☐	
CITY_AT	VARCHAR(15)	☐	☒	☐	
ZIPCODE_AT	DECIMAL(0,5)	☐	☒	☐	
STREET_AT	VARCHAR(35)	☐	☒	☐	
CLASS_AT_CODE_AT	VARCHAR(2)	☒	☒	☐	

PARENT_AT

PARENT_AT - Table					
Name: PARENT_AT					
Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒	
FIRST_NAME_AT	VARCHAR(15)	☐	☒	☐	
LAST_NAME_AT	VARCHAR(15)	☐	☒	☐	
MIDDLE_NAME_AT	VARCHAR(15)	☐	☐	☐	
HOME_PHONE_AT	CHAR(13)	☐	☐	☐	
MOBILE_PHONE_AT	CHAR(13)	☐	☒	☐	
WORK_PHONE_AT	CHAR(13)	☐	☐	☐	
EMAIL_AT	VARCHAR(30)	☐	☐	☐	
RELATION_AT	VARCHAR(20)	☐	☒	☐	

CLASS_AT

Name: CLASS_AT					
Column	Datatype	PK	NN	UQ	
CODE_AT	VARCHAR(2)	☒	☒	☒	
AGE_LEVEL_AT	VARCHAR(7)	☐	☒	☐	
COLOR_AT	VARCHAR(5)	☐	☒	☐	
ROOM_NUMBER_AT	INT	☐	☒	☐	
PHONE_NUMBER_AT	CHAR(13)	☐	☒	☐	
DIRECTION_AT	VARCHAR(160)	☐	☐	☐	
CAPACITY_AT	INT	☐	☒	☐	

BRIDGE: RELATIONSHIP_AT

- (STUDENT & PARENT)

RELATIONSHIP_AT - Table					
Name: RELATIONSHIP_AT					
Column	Datatype	PK	NN	UQ	
STUDENT_AT_ID_AT	INT	☒	☒	☐	
PARENT_AT_ID_AT	INT	☒	☒	☐	

BRIDGE: ASSIGNMENT_AT

- (CLASS & TEACHER)

ASSIGNMENT_AT - Table				
Name: ASSIGNMENT_AT				
Column	Datatype	PK	NN	
CLASS_AT_CODE_AT	VARCHAR(2)	☒	☒	
TEACHER_AT_ID_AT	INT	☒	☒	

LUNCH_ORDER_AT

Name: LUNCH_ORDER_AT					
Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒	
QUANTITY_AT	INT	☐	☒	☐	
DATE_AT	DATE	☐	☒	☐	
STUDENT_AT_ID_AT	INT	☐	☒	☐	

BRIDGE: ORDER_ITEM_AT

- (LUNCH_ORDER & LUNCH_ITEM)

Name: ORDER_ITEM_AT					
Column	Datatype	PK	NN	UQ	
LUNCH_ORDER_AT_ID_AT	INT	☒	☒	☐	
LUNCH_ITEM_AT_CODE_AT	INT	☒	☒	☐	

LUNCH_ITEM_AT

Name: LUNCH_ITEM_AT					
Column	Datatype	PK	NN	UQ	
CODE_AT	INT	☒	☒	☒	
ITEM_NAME_AT	VARCHAR(26)	☐	☒	☐	
CALORIES_AT	INT	☐	☒	☐	